

SMART LIBRARY MANAGEMENT SYSTEM

CHAPTER 1: INTRODUCTION

1.1 Introduction

The **Smart Library Management System** is a web-based application developed using the ServiceNow platform to automate and simplify library management activities. The system provides a centralized platform for maintaining book records, managing library members, issuing and returning books, calculating fines, and generating reports. It replaces traditional manual methods with a secure and efficient digital solution that improves operational efficiency and reduces human errors.

Libraries are an essential part of educational institutions because they provide students and faculty with access to learning resources. In many institutions, library operations are still managed manually using registers and spreadsheets. Manual systems require significant time and effort to maintain records and often result in duplicate entries, incorrect information, delayed updates, and difficulty in tracking book availability.

The Smart Library Management System eliminates these problems by using ServiceNow's low-code platform to automate routine tasks and maintain accurate digital records. The system provides different user roles such as Administrator, Librarian, and Student. Access to the system is controlled through Access Control Lists (ACLs), ensuring that only authorized users can perform specific operations.

Business Rules automate server-side processes, Client Scripts provide client-side validation, and UI Policies enhance the user interface by dynamically controlling field visibility and mandatory fields. The application also provides reports and dashboards that help librarians monitor book availability, issued books, member information, and fine collection. These features enable better decision-making and improve the overall efficiency of library operations.

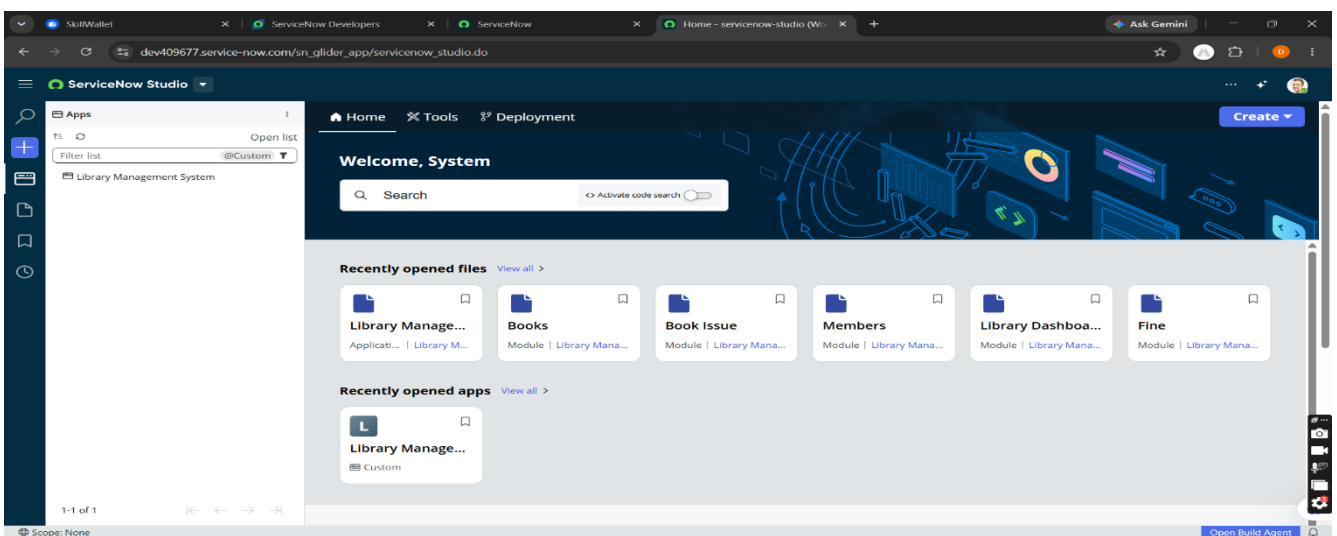


Figure 1.1: ServiceNow Scoped Application Studio Environment

1.2 Existing System

Traditional library management systems are generally manual and paper-based. Librarians maintain physical registers to record books, members, issue details, return details, and fines. Such systems are time-consuming, difficult to maintain, and prone to human errors.

Major Drawbacks of the Existing System

- Manual maintenance of book records using traditional ledgers.
- Difficulty in tracking dynamically issued and returned books.
- Increased paperwork and physical storage dependency requirements.
- Duplicate or incorrect data entries in handwritten physical logs.
- Delayed fine calculation and complex tracking protocols.
- Difficulty in compiling, sorting, and generating administrative reports.
- Poor data security and vulnerability to structural or physical damage.
- Time-consuming manual search queries for specific book details.
- Lack of a centralized database management system instance.
- No automated operational notifications or field input validations.

1.3 Proposed System

The proposed Smart Library Management System is designed to overcome the limitations of manual library management. The application automates the complete library workflow using the ServiceNow platform. The system enables librarians to manage books, maintain member information, issue and return books, calculate fines, and generate reports through an intuitive user interface. The application also provides role-based security to ensure that only authorized users can access sensitive information.

Key Features of the Proposed System

- Digital management and cloud hosting of library records.
- Comprehensive Book inventory and Member profile management modules.
- Automated Book checkout, issue, and return transaction logs.
- Automated real-time Fine calculations and status monitoring.
- Interactive and customized analytical Reports and Dashboards.
- Business Rule server-side workflow automation logic.

- Client-side validation and dynamic user-responsive UI Policies.
- Secure role-based platform access control mechanisms.

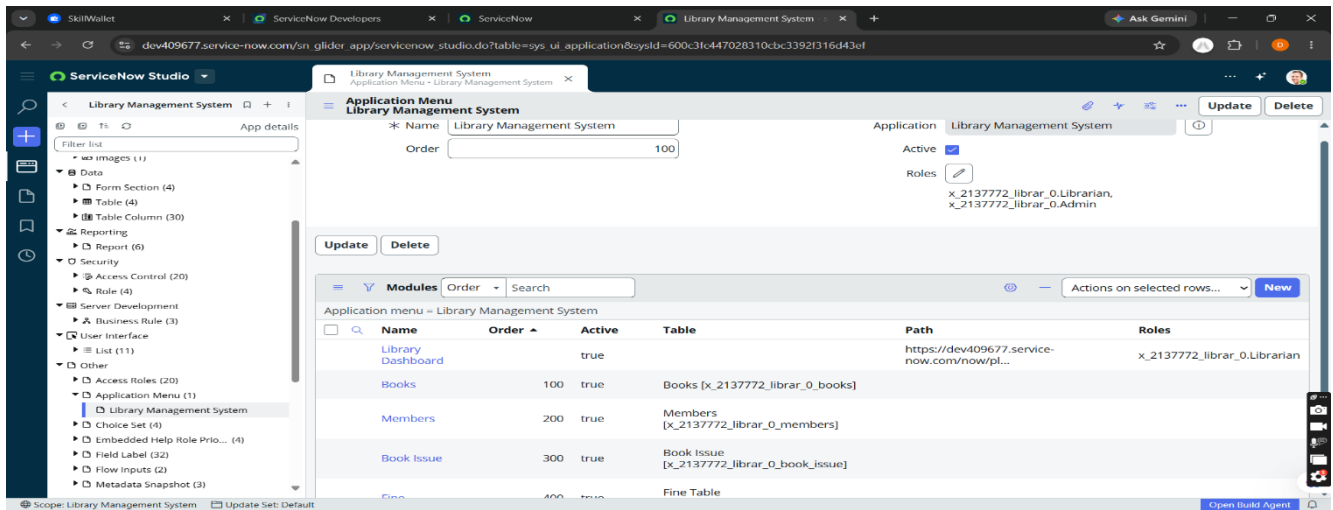


Figure 1.2: Scoped Application Menu Hierarchy in ServiceNow Navigator

1.4 Problem Statement

Many educational institutions still rely on manual library management systems that consume significant time and effort. Maintaining physical records often results in inaccurate information, duplicate entries, delayed updates, and difficulty in tracking issued books and fines. Manual systems also make report generation difficult and reduce operational efficiency.

The Smart Library Management System addresses these issues by providing an automated, centralized, and secure platform for managing library operations using ServiceNow. The system ensures accurate data management, automated workflows, secure access, and efficient reporting.

1.5 Objectives

The primary objectives of the Smart Library Management System are:

- To automate core manual library management activities completely.
- To maintain accurate, centralized records of books and registered members.
- To simplify and optimize book checkout, issue, and return protocols.
- To drastically reduce manual paperwork and eliminate redundant data entries.
- To automate financial liabilities, track outstanding tasks, and manage fine data.
- To implement strict, secure role-based access controls across modules.
- To provide robust analytical reports and visual dashboards for administration.

1.6 Scope of the Project

The Smart Library Management System can be effectively implemented in schools, colleges, universities, public libraries, and corporate organizations. The system is designed to support both librarians and students by providing secure access to library services.

The system can be further enhanced by integrating barcode scanning, QR code support, email notifications, SMS alerts, and online reservation functionality.

1.7 Advantages

- **Reduces Paperwork:** Minimizes manual documentation and physical registry requirements.
- **Improves Operational Efficiency:** Automates standard core workflows seamlessly.
- **Centralized Data Management:** Provides a single secure database platform architecture.
- **Enhances Data Security:** Protects data elements using strict role-based parameters.
- **Reduces Duplicate Records:** Eliminates human oversight errors and redundant table fields.
- **Quick Search Functionality:** Offers instant query parsing for book and member records.

CHAPTER 2: REQUIREMENT ANALYSIS

2.1 Overview

Requirement analysis identifies and documents the functional and non-functional requirements of a software system before development. It helps developers understand user expectations and ensures the application satisfies business goals. The application has been developed using the ServiceNow low-code platform to replace manual operations with an efficient digital ecosystem.

2.2 Functional Requirements

FR-1 Book Management

- The system must allow librarians to add, update, delete, and search books within the repository.
- Every book record must contain a title, author, category, publisher, ISBN number, availability status, and publication details.
- Librarians must be able to maintain and modify the active inventory of available books.

FR-2 Member Management

- The application must maintain complete information profiles about library members.
- Each member record should include: Member ID, Member Name, Email Address, Mobile Number, Department, and Membership Date.

FR-3 Book Issue Management

- The system must allow librarians to issue books to active, registered members.
- Each issue transaction must store: Member Name, Book Name, Issue Date, Due Date, Return Date, and Status.

FR-4 Fine Management

- The Fine module must calculate, record, and maintain details regarding overdue fees.
- The application must track: Fine ID, Member Name, Book Name, Fine Amount, Fine Status, and

2.3 Non-Functional Requirements

- **Performance:** The application should retrieve records quickly and respond within an acceptable time even when handling large volumes of concurrent data rows.
- **Security:** Secure access using user authentication, custom application roles, and Access Control Lists (ACLs) so only authorized users perform modifications.
- **Reliability:** Maintain precise information without risk of data loss, preserving consistency during standard CRUD platform operations.
- **Scalability:** Fully support future extensions such as barcode/QR code scanners, external communication triggers (Email/SMS), and remote online reservations.

2.4 Software & Hardware Requirements

SOFTWARE COMPONENT	PROJECT PURPOSE
ServiceNow Personal Developer Instance	Core Scoped Application Environment and Script Hosting
Web Browser (Chrome, Firefox, Safari)	Accessing ServiceNow Developer Studio cloud workspace interface
Internet Connection	Cloud Instance Platform Syncing and Remote Workspace Tools
GitHub Control Link	Source Repository Backup, Branch Isolation, and Version Controls

HARDWARE PARAMETER	MINIMUM TECHNICAL SPECIFICATION
Processor	Intel i3 / AMD Ryzen 3 or Higher performance equivalent
RAM Capacity	4 GB Minimum (8 GB Recommended for optimal browser loading)
Storage Allocation	20 GB Free Physical Disk Space minimum requirement
Monitor Resolution	1366 × 768 Pixels standard layout resolution

2.5 Execution Roadmap

- **Phase 1:** System Requirement Analysis → Conceptual Planning → Relational Database Design.
- **Phase 2:** Table Instantiation → Field Configuration → User/Role Generation → ACL Policy Mapping.
- **Phase 3:** Implementation of Server Business Rules, Client Scripts, Dynamic UI Policies, Reports, and Dashboards.
- **Phase 4:** Functional Testing Analysis → Technical Documentation → Cloud Deployment → GitHub Repository Sync.

CHAPTER 3: SYSTEM DESIGN

3.1 Overview

System design defines the architecture, database structure, workflow, and overall organization of the application. It provides a blueprint of how different components interact to achieve system goals using a role-based security structure.

3.2 System Architecture

The Smart Library Management System follows a layered architecture structured as follows:

- **User Layer:** End-users (Administrators, Librarians, Students) accessing the application through the web interface.
- **Application Layer:** Contains the business logic, including Business Rules, Client Scripts, UI Policies, Flow Designer, Reports, and Dashboards.
- **Database Layer:** Manages application data stored in relational tables (Books, Members, Book Issue, Fine) connected via reference fields.
- **Security Layer:** Enforces data protection using User Authentication, Roles, and Access Control Lists (ACLs) to manage CRUD operations.

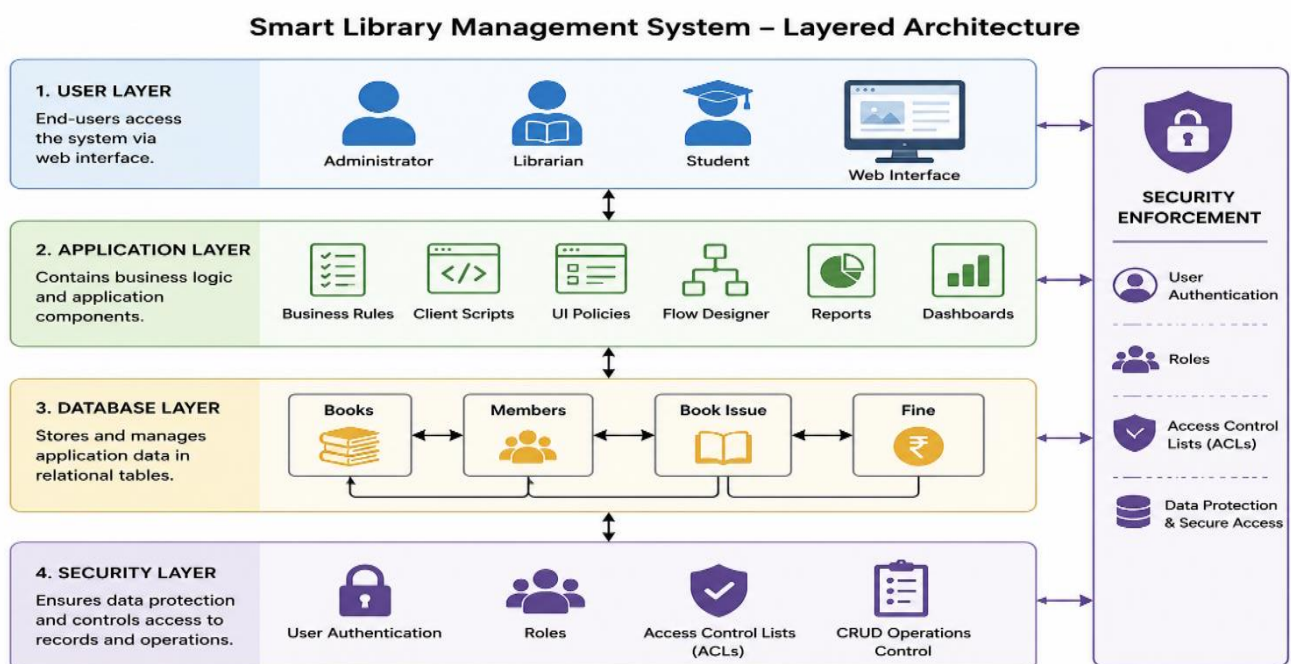


Figure 3.1: Layered Architecture Block Diagram Layout

3.3 Entity Relationship (ER) & Database Design

The relational schema ensures data integrity and eliminates redundancy. The logical relational paths mapping the schema can be represented textually as:

Books → Referenced by → Book Issue ← Referenced by ← Members

Book Issue → Referenced by → Fine

Smart Library Management System – ER Diagram

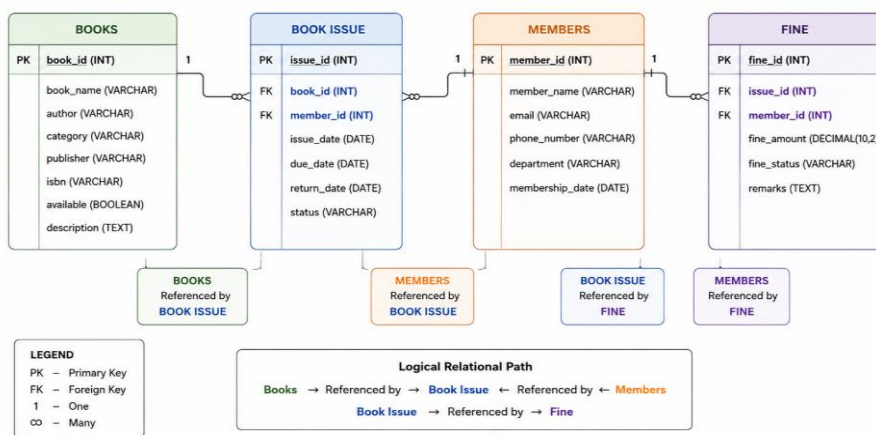


Figure 3.2: Entity Relationship Diagram Schema Model

Table Architecture Parameters

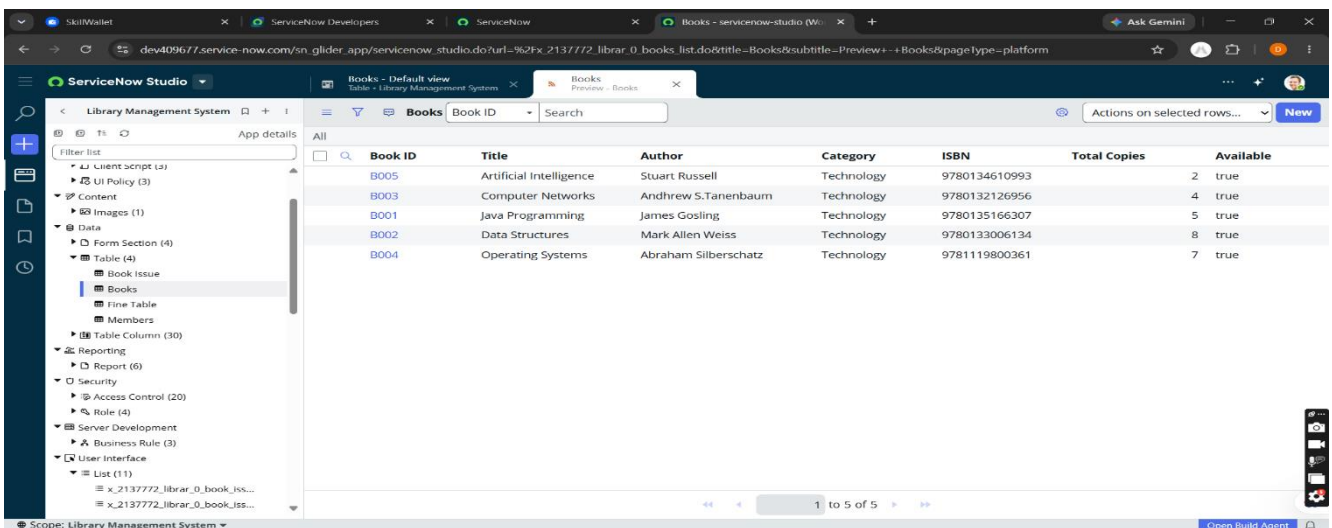
- **Books Table:** Book ID, Book Name, Author, Category, Publisher, ISBN, Available (True/False), Description.
- **Members Table:** Member ID, Member Name, Email, Phone Number, Department, Membership Date.
- **Book Issue Table:** Issue ID, Book (Reference), Member (Reference), Issue Date, Due Date, Return Date, Status.
- **Fine Table:** Fine ID, Book Issue (Reference), Member (Reference), Fine Amount, Fine Status, Remarks.

CHAPTER 4: IMPLEMENTATION

4.1 Implementation Overview

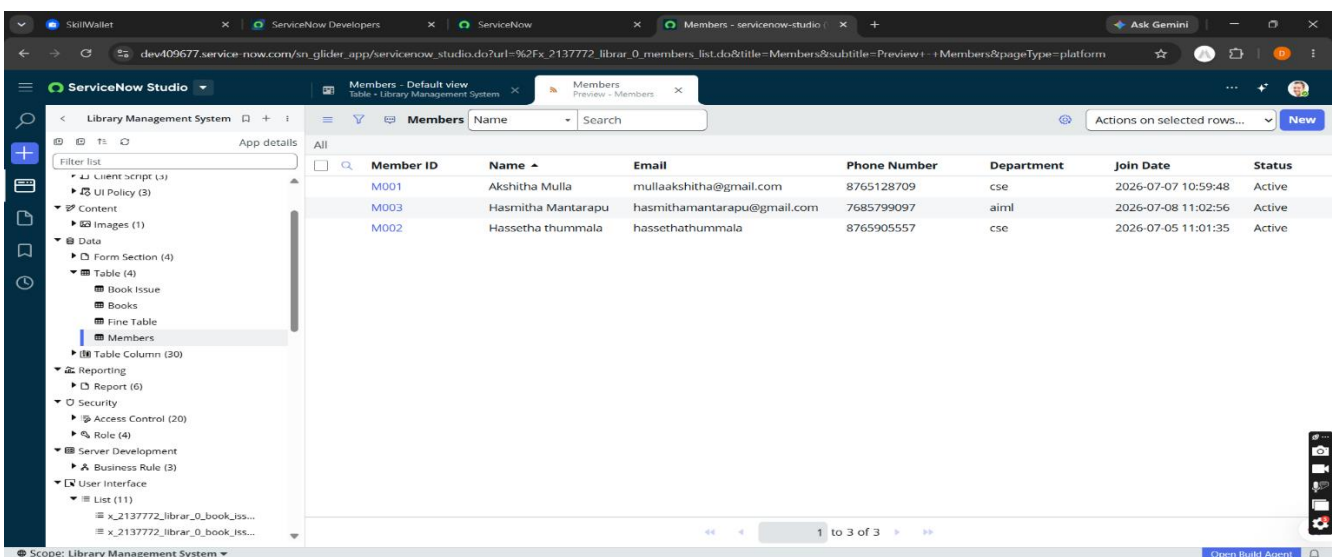
The application was built as a secure **Scoped Application** inside *ServiceNow Studio*. All data schemas, access controls, automated business workflows, and user views were compiled into this independent namespace to ensure deployment reliability.

4.2 Module Setup and Form Fields



Book ID	Title	Author	Category	ISBN	Total Copies	Available
B005	Artificial Intelligence	Stuart Russell	Technology	9780134610993	2	true
B003	Computer Networks	Andrew S.Tanenbaum	Technology	9780132126956	4	true
B001	Java Programming	James Gosling	Technology	9780135166307	5	true
B002	Data Structures	Mark Allen Weiss	Technology	9780133006134	8	true
B004	Operating Systems	Abraham Silberschatz	Technology	9781119800361	7	true

Figure 4.1: Books Table Custom Dictionary and List Interface



Member ID	Name	Email	Phone Number	Department	Join Date	Status
M001	Akshitha Mulla	mullaakshitha@gmail.com	8765128709	cse	2026-07-07 10:59:48	Active
M003	Hasmitha Mantarapu	hasmithamantarapu@gmail.com	7685799097	aiml	2026-07-08 11:02:56	Active
M002	Hassetha thummala	hassethathummala	8765905557	cse	2026-07-05 11:01:35	Active

Figure 4.2: Members Table Registration Schema Interface

Issue ID	Book	Member	Issue Date	Due Date	Return Date	Status
1002	Artificial Intelligence	Hasmitha Mantarapu	2026-07-08 05:50:30	2026-07-22 05:50:45	(empty)	Issued
1003	Java Programming	Hassetha thummala	2026-07-05 05:51:51	2026-07-19 05:52:08	2026-07-07 05:52:31	Returned
1001	Operating Systems	Akshitha Mulla	2026-07-09 12:29:16	2026-07-23 05:48:44	(empty)	Issued

Figure 4.3: Book Issue Transaction Management Table

Fine ID	Book Issue	Fine Amount	Paid	Payment Date
F001	Akshitha Mulla	0	true	2026-07-23 05:56:23
F003	Hassetha thummala	100	true	2026-07-18 05:58:28
F002	Hasmitha Mantarapu	50	false	2026-07-22 05:57:33

Figure 4.4: Fine Computation Logs and Accounting Records

4.1 Scripting & Automation Logic

Business Rules (Server-Side Automation)

Business Rules handle backend data integrity checks. When a book transaction state changes, server-side logic dynamically sets the target book record's availability flag (*true / false*) and runs database validity procedures to prevent concurrency anomalies.

Name	Active	Table	Application	Order	Updated
Book Available on Return	true	Book Issue [x_2137772_librar_0_book_issue]	Library Management System	100	2026-07-11 10:28:20
Book Unavailable on Issue	true	Book Issue [x_2137772_librar_0_book_issue]	Library Management System	100	2026-07-11 10:27:11
Prevent Duplicate Book Issue	true	Book Issue [x_2137772_librar_0_book_issue]	Library Management System	100	2026-07-11 10:29:05

Figure 4.5: Server-Side Business Rule Configuration Script Editor

Client Scripts (Client-Side Validations)

UI Policies (Dynamic User Interface)

UI Policies dynamically orchestrate form layouts. Based on form data state inputs, fields dynamically transition into mandatory, read-only, or hidden states without writing extensive client script configurations.

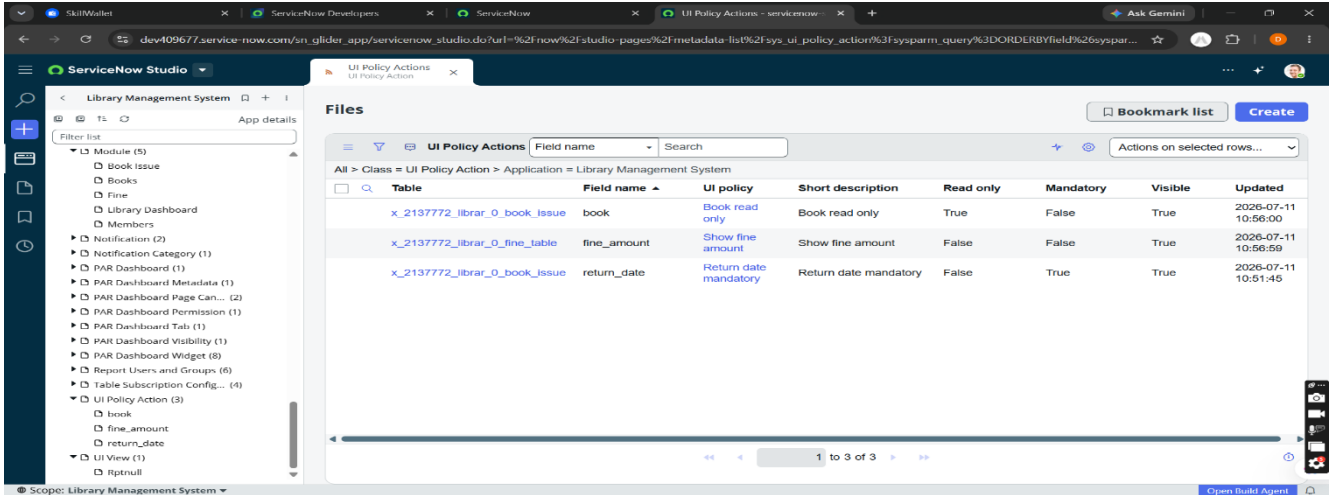
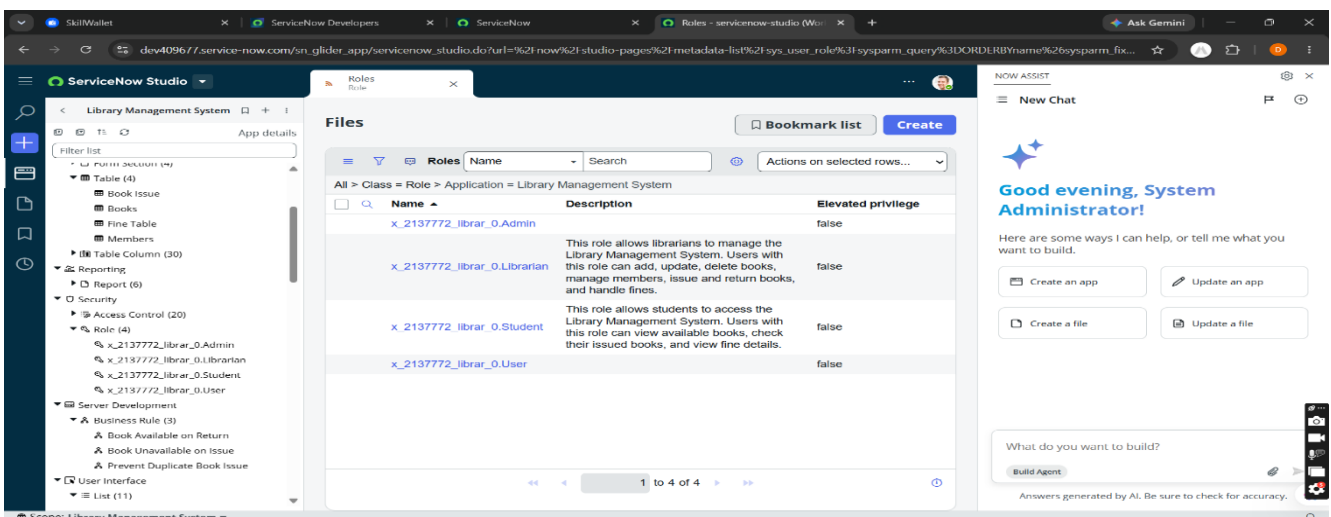


Figure 4.7: UI Policy Operations Conditions Dashboard

4.2 Role-Based Access Controls (ACLs) & Users

Granular platform access control lists (ACLs) safeguard data operations across administrative, operations, and consumer layers:

- **Administrator Role:** Holds master rights to adjust underlying infrastructure, configure access policies, and manage core table mappings.
- **Librarian Role:** Retains CRUD permissions across system transaction records to perform daily library management duties.
- **Student Role:** Holds read-only access flags allowing execution of search indexes and checking status metrics.



The screenshot shows the ServiceNow 'Users' list configuration page. The table displays the following data:

User ID	Name	Email	Active	Created	Updated
0	Librarian	librarian@gmail.com	true	2026-07-09 23:18:45	2026-07-09 23:18:45
1	Akshitha Mulla	akshithamulla@gmail.com	true	2026-07-09 23:15:21	2026-07-09 23:15:21
2	Hasmitha Mantharapu	hasmithamantharapu@gmail.com	true	2026-07-09 23:16:11	2026-07-09 23:16:11
3	Haseetha Thummala	haseethathummala@gmail.com	true	2026-07-09 23:17:43	2026-07-09 23:17:43
abel.tuter	Abel Tuter	abel.tuter@example.com	true	2012-02-17 19:04:52	2026-07-07 21:35:49
abraham.lincoln	Abraham Lincoln	abraham.lincoln@example.com	true	2013-07-23 17:15:54	2026-07-07 21:35:52
adela.cervantsz	Adela Cervantsz	adela.cervantsz@example.com	true	2012-02-17 19:04:50	2026-07-07 21:35:46
admin	System Administrator	admin@example.com	true	2007-07-03 11:48:47	2026-07-08 01:15:11
aes.creator	Creator User		true	2026-07-07 22:30:15	2026-07-08 01:15:11
aileen.mottern	Aileen Mottern	aileen.mottern@example.com	true	2012-02-17 19:04:49	2026-07-07 21:35:49
alejandra.prenatt	Alejandra Prenatt	alejandra.prenatt@example.com	true	2012-02-17 19:04:52	2026-07-07 21:35:47
alejandro.mascall	Alejandro Mascall	alejandro.mascall@example.com	true	2012-02-17 19:04:52	2026-07-07 21:35:52
alene.rabeck	Alene Rabeck	alene.rabeck@example.com	true	2012-02-17 19:04:53	2026-07-07 21:35:53
alfonso.griglen	Alfonso Griglen	alfonso.griglen@example.com	true	2012-02-17 19:04:51	2026-07-07 21:35:46
alissa.mountjoy	Alissa Mountjoy	alissa.mountjoy@example.com	true	2012-02-17 19:04:52	2026-07-07 21:35:49
allan.schwandt	Allan Schwandt	allan.schwandt@example.com	true	2012-02-17 19:04:53	2026-07-07 21:35:52

Figure 4.9: Registered Library Users Profile Configuration Logs

The screenshot shows the ServiceNow Studio 'Access Controls' configuration page for the 'Library Management System'. The table displays the following data:

Name	Decision Type	Operation	Type	Active	Updated by	Updated
x_2137772_librar_0_books	Allow If	read	record	true	admin	2026-07-11 10:18:46
x_2137772_librar_0_books	Allow If	delete	record	true	admin	2026-07-11 10:20:52
x_2137772_librar_0_books	Allow If	create	record	true	admin	2026-07-11 10:20:18
x_2137772_librar_0_books	Allow If	write	record	true	admin	2026-07-11 10:22:21
x_2137772_librar_0_books	Allow If	read	record	true	admin	2026-07-11 10:19:48
x_2137772_librar_0_book_issue	Allow If	write	record	true	admin	2026-07-11 09:52:25
x_2137772_librar_0_book_issue	Allow If	delete	record	true	admin	2026-07-11 09:53:01
x_2137772_librar_0_book_issue	Allow If	create	record	true	admin	2026-07-11 09:51:45
x_2137772_librar_0_book_issue	Allow If	read	record	true	admin	2026-07-11 09:50:23
x_2137772_librar_0_book_issue	Allow If	read	record	true	admin	2026-07-11 09:53:46
x_2137772_librar_0_fine_table	Allow If	read	record	true	admin	2026-07-11 10:03:59
x_2137772_librar_0_fine_table	Allow If	delete	record	true	admin	2026-07-11 10:18:12
x_2137772_librar_0_fine_table	Allow If	read	record	true	admin	2026-07-11 10:16:24

Figure 4.10: Table and Field-Level Security Access Control Lists (ACLs)

CHAPTER 5: TESTING, RESULTS AND CONCLUSION

5.1 System Testing Methodology

Comprehensive testing verified the functional accuracy, platform execution metrics, and security controls across all configuration fields.

5.2 Functional Testing Analysis

TEST CASE ID	TEST TARGET COMPONENT	EXPECTED RESULT OUTCOME	STATUS FLAG
TC-FR-01	Create Book Record	Book entry inserts and saves perfectly	Pass
TC-FR-02	Update Book Record	Field corrections sync across table rows	Pass
TC-FR-03	Delete Book Record	Target row purges clean from active database	Pass
TC-FR-04	Register New Member	Member metadata maps successfully to records	Pass
TC-FR-05	Book Issue Protocol	Transaction record generates with matching constraints	Pass
TC-FR-06	Book Return System	Status labels switch to return values	Pass
TC-FR-07	Server Business Rules	Triggers calculate automatically upon database calls	Pass
TC-FR-08	Interface Client Scripts	Form validation flags throw errors immediately	Pass

5.3 Analytics, Reports and Dashboards

Visual analytics elements track execution data real-time, displaying structural status details directly onto the primary application workspace views.

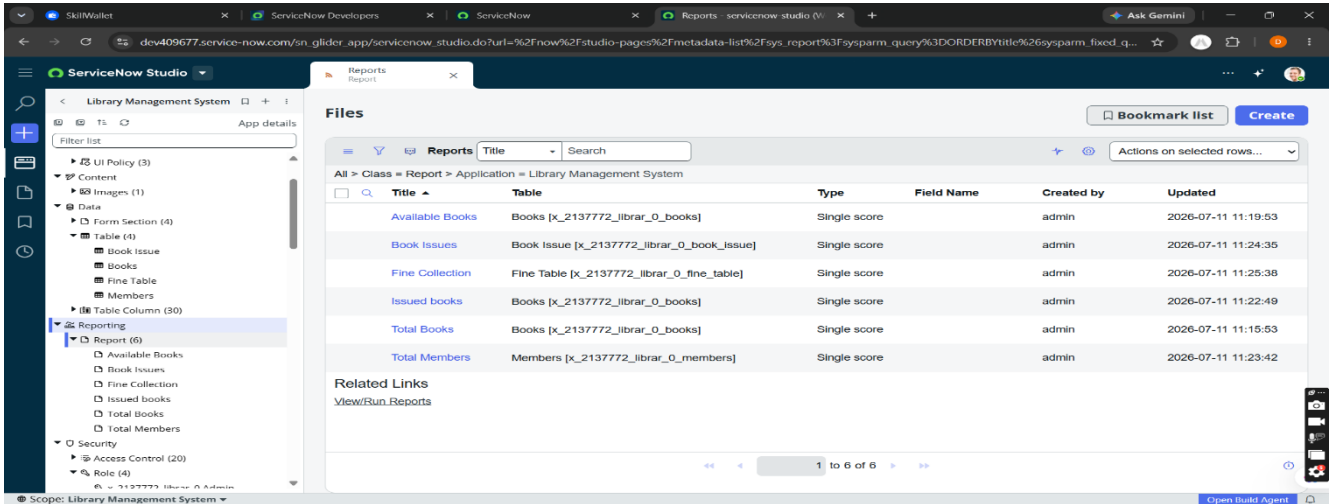


Figure 5.1: Live System Performance Reports Generated Modules

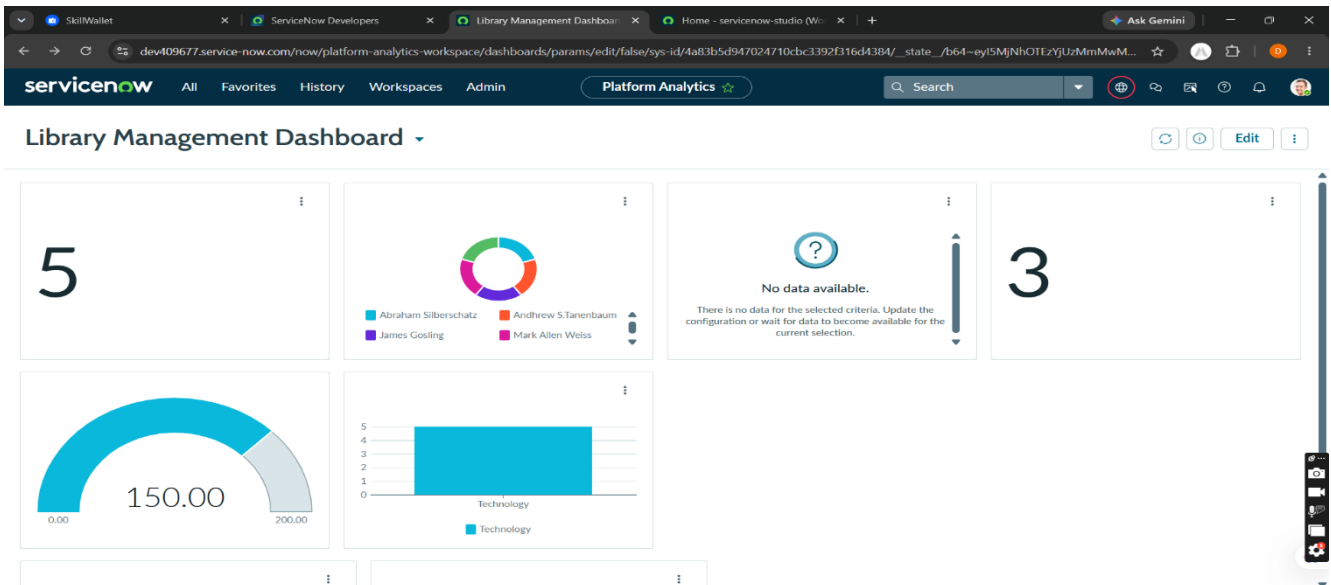


Figure 5.2: Central Management Interactive Admin Dashboard Panel

5.4 Conclusion

The Smart Library Management System was successfully developed and deployed on the ServiceNow cloud environment. By utilizing platform-native features like Scoped Applications, automated Business Rules, dynamic UI Policies, and custom analytics widgets, the application provides a modern solution for running day-to-day library operations efficiently. The system replaces manual data entry tasks, prevents record duplication, secures data transactions, and scales cleanly to fit institutional growth needs.